

Evaluation Methodology for GNN Explainer Quality Metrics

1 Introduction

We evaluate GNNExplainer explanations using three metrics: fidelity, unfaithfulness, and sparsity. Two of PyG’s built-in metrics—fidelity and sparsity—are unsuitable for our R-GCN wrapper architecture and variable-size molecular graphs, so we propose custom formulations. We retain the built-in unfaithfulness metric unchanged.

2 Probability-Based Fidelity

2.1 Why the Built-in Fidelity Fails

The PyG `fidelity()` function [1] tests whether the predicted class changes when the important subgraph is removed (fid_+) or retained alone (fid_-):

$$\text{fid}_+ = 1 - \frac{1}{N} \sum_{i=1}^N \mathbb{K}(\hat{y}_i^{G_{C \setminus S}} = \hat{y}_i), \quad \text{fid}_- = 1 - \frac{1}{N} \sum_{i=1}^N \mathbb{K}(\hat{y}_i^{G_S} = \hat{y}_i) \quad (1)$$

This fails in our setting for two reasons. First, it uses `set_masks()` to apply edge masks via forward hooks on message-passing layers, but our `RGCNExplainerWrapper` delegates to an inner `MutagenicityGNN` containing the `FastRGCNConv` layers. The hooks register on the wrapper, not the inner model, so masks have *no effect*—yielding degenerate $\text{fid}_+ = 1$, $\text{fid}_- = 0$ for every graph. Second, the binary $\mathbb{K}(\cdot)$ indicator produces only $\{0, 1\}$ per graph, providing no information about the *degree* of influence on model confidence.

2.2 Custom Probability-Drop Formulation

Let $p_c(G) = \text{softmax}(f_\theta(G))_c$ be the probability assigned to the originally predicted class c . We explicitly mask node features and prune edges before the forward pass, then measure the probability drop:

$$\boxed{\text{fid}_+ = \max(p_c(G) - p_c(G_{C \setminus S}), 0)} \quad (2)$$

$$\boxed{\text{fid}_- = \max(p_c(G) - p_c(G_S), 0)} \quad (3)$$

For fid_+ , we remove the important subgraph (multiply node features by $1 - \mathbf{m}_{\text{node}}$, prune edges with $\text{mask} > 0.5$). For fid_- , we keep only the important subgraph (multiply by $\mathbf{m}_{\text{node}}$, prune edges with $\text{mask} \leq 0.5$). This yields continuous values in $[0, 1]$, is sensitive to confidence changes even without class flips, and bypasses the hook-propagation bug entirely.

3 Unfaithfulness

We retain the PyG built-in unfaithfulness metric [2], which measures the KL divergence between the original and mask-reproduced output distributions:

$$\text{UF} = D_{\text{KL}}(\mathbf{p} \parallel \mathbf{p}^M) \quad (4)$$

Lower values indicate the explanation faithfully reproduces model behaviour. The mask-hook bug does not affect this metric because it is computed during the explainer’s internal optimisation where hooks target the base model correctly.

4 Topology-Aware Sparsity

4.1 Why the Built-in Sparsity Fails

The standard sparsity—fraction of edges below a threshold ϵ —has two problems: (1) threshold ϵ is arbitrary on continuous $[0, 1]$ masks with no natural cutoff, and (2) it ignores graph-size variation. In MUTAG, molecules range from ~ 10 to ~ 40 nodes; a fixed threshold inflates sparsity for large graphs and deflates it for small ones, making scores incomparable.

4.2 Custom Topology-Aware Formulation

Given edge mask $\mathbf{m}_e \in [0, 1]^{|E|}$ on graph $G = (V, E)$:

Step 1. Select the top- k most important edges:

$$E_{\text{top}} = \arg \max_{E' \subseteq E, |E'|=k} \sum_{e \in E'} m_e \quad (5)$$

Step 2. Identify incident nodes:

$$V_{\text{top}} = \{v \in V : \exists e = (u, v) \in E_{\text{top}}\} \quad (6)$$

Step 3. Compute sparsity as the fraction of graph elements not in the explanation:

$$\boxed{\text{sparsity} = 1 - \frac{|V_{\text{top}}| + |E_{\text{top}}|}{|V| + |E|}} \quad (7)$$

This is size-invariant (normalised by $|V| + |E|$ per graph), threshold-free (top- k has a clear interpretation), and topology-aware (incident nodes count reflects structural coherence).

5 Results

We evaluate all metrics on the MUTAG test set over 56 correctly classified molecules using a trained R-GCN with GNNExplainer (200 epochs, lr=0.01). Results are shown in Table 1.

Metric	Value	Desired
Fidelity ₊	0.6875	High
Fidelity ₋	0.0164	Low
Unfaithfulness	0.0500	Low
Sparsity	0.6388	High

Table 1: Average explanation quality over 56 correctly classified test molecules.

Interpretation. A fid_+ of 0.6875 indicates that removing the explained subgraph drops the predicted-class probability by 68.75% on average, confirming the explainer identifies genuinely important structure. A low fid_- of 0.0164 means the important subgraph alone nearly preserves the original prediction, indicating the explanation is sufficiently complete. Low unfaithfulness (0.05) shows the explanation masks closely reproduce the model’s output distribution. A sparsity of 0.64 means the top-10 important edges (and their incident nodes) constitute only 36% of each graph on average, confirming explanations are concise rather than trivially covering the full input.

References

- [1] F. Faber, A. K. Mishra, S. K. Parida, et al., “GraphFramEx: Towards Systematic Evaluation of Explainability Methods for Graph Neural Networks,” *arXiv preprint arXiv:2206.09677*, 2022.
- [2] R. Ying, D. Bourgeois, J. You, M. Zitnik, and J. Leskovec, “GNNExplainer: Generating Explanations for Graph Neural Networks,” *Advances in Neural Information Processing Systems (NeurIPS)*, 32, 2019.